

Quang HOANG

Software Engineer Apprentice @ Bouygues Telecom

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🌐 Portfolio 🐙 Github 🔗 LinkedIn

PROFESSIONAL EXPERIENCE

Software Engineer Apprentice | Bouygues Telecom

Sept. 2024 – Sept. 2026

Reference : Thierry DELACOUX | tdelacou@bouyguetelecom.fr

AWS PostgreSQL Redis Java JanusGraph RabbitMQ Ansible

- **High-Scale Distributed Architecture** : Designed a fault-tolerant platform managing telecom network metadata (hundreds of thousands of nodes), powered by **highly-concurrent Java microservices** and RabbitMQ to decouple massive data flows.
- **Low-Latency Caching** : Implemented a distributed storage architecture (JanusGraph, ElasticSearch) integrated with **Redis caching**, ensuring high consistency and **reducing query latency** for complex multi-hop topology reads by **over 90%** compared to legacy relational queries.
- **Database Optimization** : Resolved PostgreSQL scalability bottlenecks under high-throughput loads by **analyzing query execution plans, optimizing compound indexes, and tuning HikariCP connection pools** to maximize resource utilization and prevent connection exhaustion.
- **AI Industrialization**: Proposed and led the **integration of a new AI-based feature**, inspired by my master's thesis on GraphRAG, thereby bridging the gap between applied research in LLMs and production platform engineering.

Backend Software Engineer Apprentice | BNP Paribas

Sept. 2022 – Sept. 2023

Reference : Ahmed Amine BALTI | amineahmed.balti@neosoft.fr

C# .Net SQL Server Multi-threading PowerShell

- **Automated Lifecycle Engine** : Developed a **multi-threaded background processing engine** in C# for internal infrastructure governance, enabling secure, parallelized decommissioning and **reducing technical debt by 30%**.
- **Operational Reliability** : Engineered automated audit jobs and proactive alerting systems, significantly improving system observability and **decreasing support tickets by 40%**.

TECHNICAL PROJECTS

Operational Intelligence - Agentic GraphRAG System

Neo4j LangGraph Python ChromaDB Docker

- **GraphRAG & LLM Architecture** : Designed a pipeline (Neo4j, LangGraph) with self-correcting Cypher generation to analyze telecom outage impact (blast radius) on multi-hop network topologies.
- **Data Engineering & Simulation** : Developed a synthetic digital twin to generate complex testing environments.
- **Evaluation (RAGAS)** : Empirical proof of system superiority over Vector RAG (ChromaDB): maintained **100% context precision** at over 3 hops, eliminating structural hallucinations.

Distributed Real-Time Chat System

TypeScript/Node.js Redis RabbitMQ PostgreSQL

- **Asynchronous Event Distribution** : Designed a containerized **Node.js (NestJS)** backend leveraging WebSockets and RabbitMQ for **highly-parallel, low-latency message distribution** across active nodes.
- **Data Access Optimization** : Mitigated "N+1" query performance bottlenecks using batched caching mechanisms (DataLoader/Redis).
- **Load Resilience** : Validated API scalability and operational readiness through rigorous **concurrency performance testing** (Artillery).

EDUCATION

Master's in Software Engineering | EFREI (Paris, France)

2024 – 2026

Bachelor's in Computer Science (DANT) | Sorbonne University

2021 – 2023

Valedictorian (Ranked 2/20)

TECHNICAL SKILLS

Languages : TypeScript, Node.js (NestJS, Express), Python, Java, Go, C#, SQL.

Data & Cloud : AWS, Kubernetes, Docker, Terraform, Kafka, RabbitMQ, GraphQL, REST APIs.

Databases : PostgreSQL, Redis, MongoDB, ElasticSearch, JanusGraph, Neo4j, ScyllaDB.

AI / MLOps : LangGraph, LangChain, GraphRAG, LLM Evaluation (RAGAS), LangFuse, MLflow.

Front-end : React, Next.js, Angular, Tailwind CSS.

INTERESTS

- **Music** : Classical guitar (Third prize in municipal competition), private piano lessons.
- **Chess** : Elo rating 1200, active participation in official over-the-board (OTB) tournaments.
- **Badminton** : Club practice (Official rankings: P10 in singles, D9 in doubles).